TCEX TECHNOLOGIES · CHENNAI

TraceCore

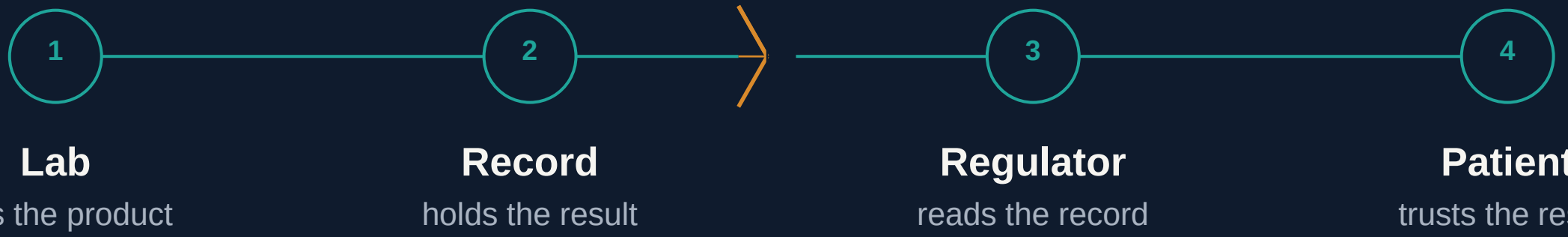
Lab records that stay trustworthy in the quantum era

Amaravati Quantum Valley stakeholder meeting, 15 October 2026

[Presenter name, role]

THE STAKE

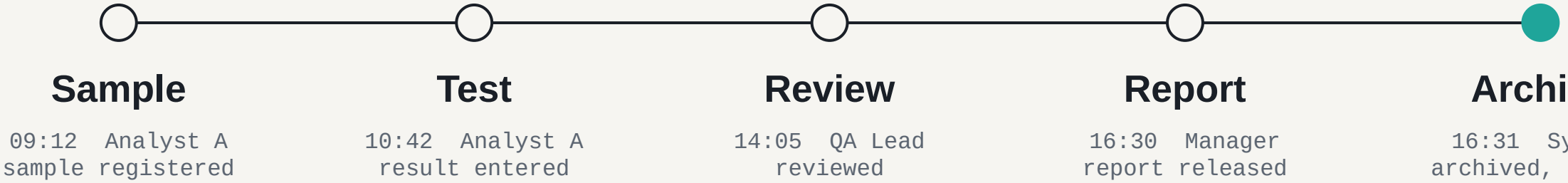
Every approved medicine and medical device rests on a lab record



If that record is altered, or leaked, the whole chain breaks.

WHAT A LIMS IS

A LIMS is the lab's system of record



Every step is logged: who did it, what changed, when, and why.

Used by pharma, medical device, CRO and food testing labs.

The question every auditor asks

“Can you prove this record was never altered?”

THE PROBLEM TODAY

Paper and spreadsheets cannot prove a record is untouched

- Records sit across paper registers, Excel files and email
- Values are re-typed by hand from instruments
- Edits leave no trace of who changed what
- Audits take days of reconstruction

22

Indian sites received US FDA warning letters in 2026, at least.

About half cited data integrity or incomplete lab records.

Source: our 1 Oct 2026 review of published US FDA warning letters.

HARVEST NOW, DECRYPT LATER

Data stolen today can be read tomorrow

Anything that must stay secret for years is already at risk.



Today

Attackers copy encrypted data and store it

Years pass

The copy sits untouched, waiting

Quantum arrives

Public-key encryption (RSA, ECC) breaks, and the data opens

THE RULE

secrecy period + migration time > time until quantum → you are already late

WHY LAB DATA IS A PRIME TARGET

Lab data must stay secret and trusted for decades

DATA	WHY IT STAYS VALUABLE
Study protocols and research data	Intellectual property for the life of the product
Sponsor and client information	Commercial contracts and unreleased products
Regulated reports	Kept for years under GLP and GMP retention rules
Signing and identity keys	They vouch for every record in the archive

QUANTUM RISK IS ALSO A FORGERY RISK

A broken signature means a record can no longer prove anything

- Today, digital signatures (RSA, ECDSA) prove who approved a record
- A capable quantum computer could forge those signatures
- An archived approval would then prove nothing
- For regulated labs, trust in the record matters as much as secrecy



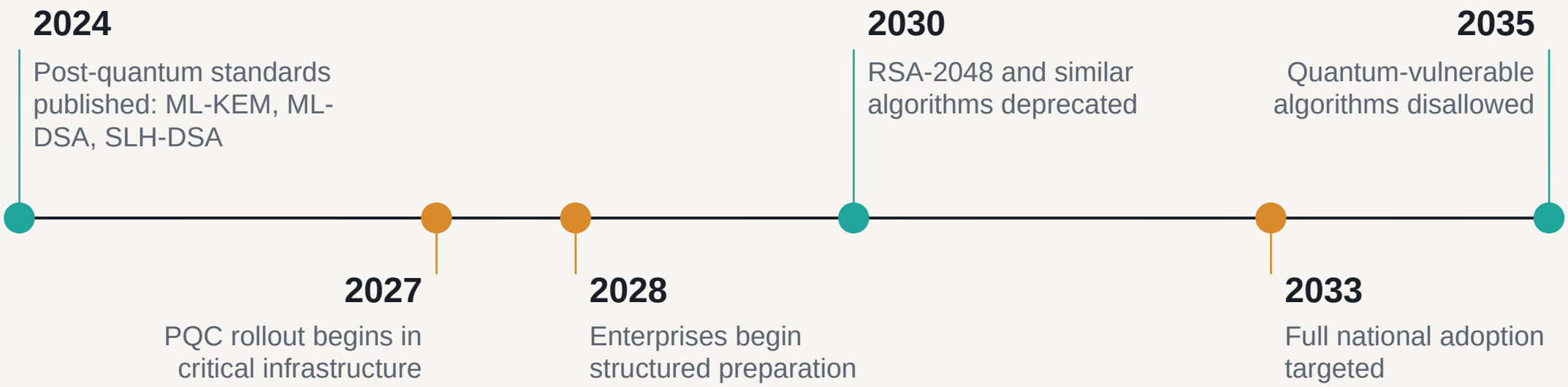
Genuine seal

Forged seal

THE MIGRATION CLOCK HAS STARTED

Standards and national deadlines are already set

NIST



INDIA

India's task force calls for crypto inventories, vendor crypto bills of materials and quantum-safe procurement.

Source: NIST IR 8547 (draft); India National Quantum Mission task force report, Feb 2026.

OUR APPROACH

TraceCore builds post-quantum protection into the lab record

Signatures and audit trail	Approvals and every logged change (ML-DSA)	[CONFIRM]
Storage and keys	Stored records and encryption keys	[CONFIRM]
Transport	Data moving between lab, instruments and server (ML-KEM)	[CONFIRM]
Access	Passwordless sign-in for each user	[CONFIRM]

STATUS KEY

Built

Tested

In progress

Planned

Crypto-agile: algorithms can be swapped without rebuilding the system [CONFIRM]

Deployed on the lab's own servers: the lab keeps its data and its keys [CONFIRM]

WHAT THE PLATFORM DOES

One connected record from sample to archive

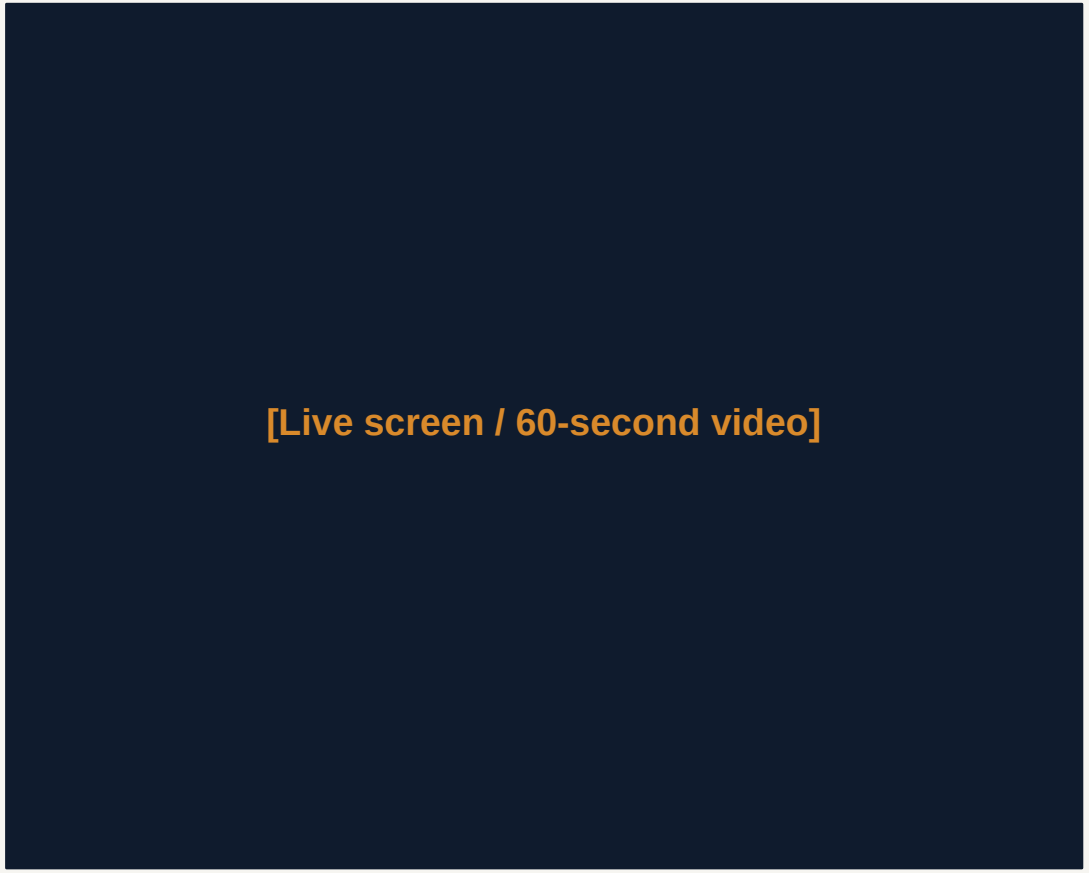
Samples	registration, unique ID, storage location, status
Studies	protocol, study director, sponsor, lifecycle
Testing	instrument data captured directly, limits checked **[CONFIRM which instruments]**
Quality	deviations, CAPA, approvals
Reports	version control, controlled release, retrieval
Integrity	role-based access, server-set timestamps, a reason for every change, original value always kept

[Insert real audit-trail screenshot]

LIVE DEMO

One sample, end to end

1. Register a sample. The server sets the date; it cannot be changed
2. Capture a result from an instrument
3. Try to edit it. The system demands a reason
4. Open the audit trail: old value, new value, who, when, why
5. Approve with an electronic signature **[show the PQC signature if built]**



HOW A LAB SHOULD JUDGE ANY LIMS

Five questions every lab should ask its software vendor

QUESTION	TRACECORE'S ANSWER
Can every record be traced from sample to archive?	Yes, linked end to end
Can I see who changed what, why, and who approved it?	Yes, full audit trail
Are roles and duties separated?	Yes, role-based access
Where does my data live, and who holds the keys?	On the lab's servers, with the lab **[CONFIRM]**
How will the system move to post-quantum cryptography?	Built in **[CONFIRM status]**

WHERE WE START

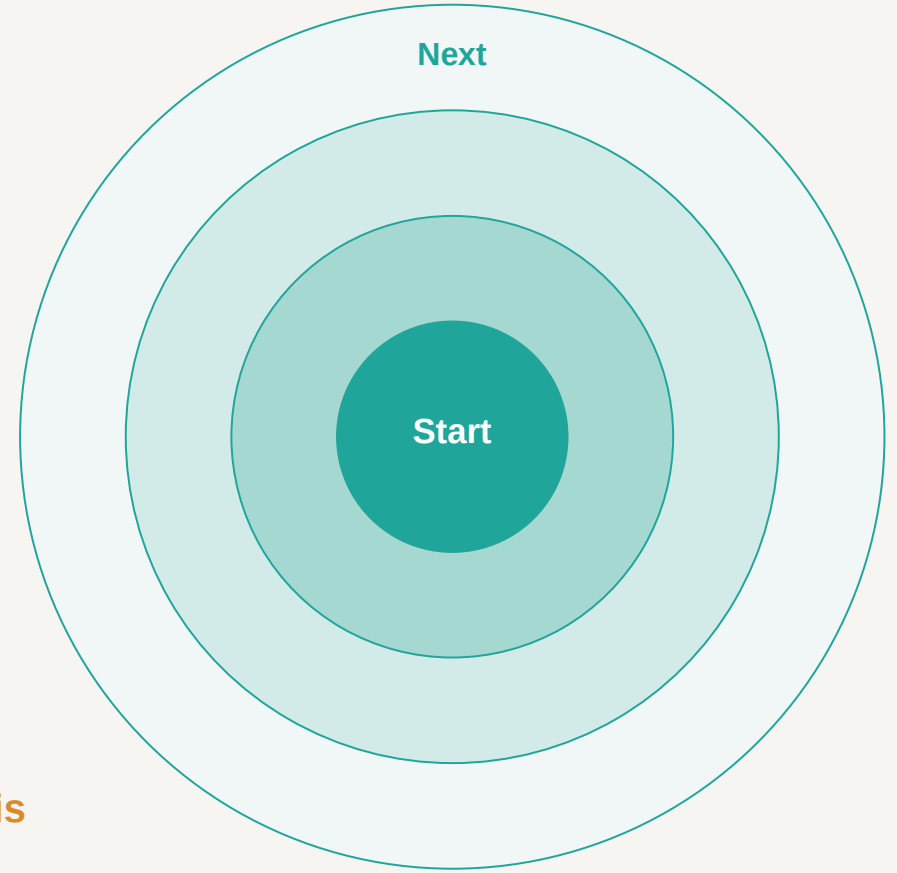
We start where data must stay protected the longest

61 certified GLP pre-clinical facilities

~89 medical device testing labs registered with CDSCO

~10,500 pharma manufacturing units, each with a QC lab

6,815 NABL-accredited testing labs (next)



First deployment: **[MDR, a leading Chennai device-testing lab, if naming is cleared]**

Source: DST GLP list, CDSCO testing-lab register, NABL portal (read 1 Oct 2026); pharma unit count is an industry estimate.

WHY THIS BELONGS IN QUANTUM VALLEY

Quantum Valley builds the compute. We protect the data it could expose.

QUANTUM COMPUTE

Amaravati

- IBM Quantum System Two, 156-qubit Heron processor
- Pharma among its target applications

QUANTUM-SAFE DATA

TraceCore

- The same pharma and device sector holds data quantum could expose
- An early reference application for quantum-safe procurement

Built in South India, for Indian regulated labs.

THE ASK

What we are asking for

01

Access to Quantum Valley testbeds to benchmark our post-quantum layer

02

Expert review of our cryptographic design by ecosystem partners

03

Introductions to 2 or 3 pilot labs in Andhra Pradesh

04

[CONFIRM: incubation, grant or funding ask, with amount]

Regulated science runs on trust. We are making that trust last.

APPENDIX A1 · REGULATORY CONTEXT

The rules lab records live under

OECD GLP	Pre-clinical study conduct and records
21 CFR Part 11	Electronic records and signatures for US FDA submissions
ALCOA+	Attributable, legible, contemporaneous, original, accurate, plus complete, consistent, enduring, available
Revised Schedule M	Indian GMP: a record of every data change, in force for all units since 1 Jan 2026

No body certifies LIMS software. Labs validate the configured system, and vendors supply validation documents.

APPENDIX A2 · STATUS OF EVERY CLAIM

Where each capability stands today

CAPABILITY	BUILT	TESTED	IN PROGRESS	PLANNED	EXT. REVIEWED
Audit trail and controlled correction	☐	☐	☐	☐	☐
Electronic signatures	☐	☐	☐	☐	☐
ML-KEM key establishment	☐	☐	☐	☐	☐
ML-DSA signatures	☐	☐	☐	☐	☐
Passwordless sign-in (KratosId)	☐	☐	☐	☐	☐
Instrument integration	☐	☐	☐	☐	☐
AI compliance assistant	☐	☐	☐	☐	☐